MAJOR PROJECT · PHASE 2 PROGRESS PRESENTATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AgriAI 360

Beyond Crop Recommendation — Yield, Profit, Water, Risk, Disease, Fertilizer, Market Price & Notification Intelligence

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TEAM MEMBERS & PRESENTATION SPLIT

Alden Pais — NNM23CS020 (Yield & Profit Prediction)

Disha Shetty — NNM23CS067 (Water Requirement & Yield Risk)

Divya R — NNM23CS068 (Disease Detection & Fertilizer Advisor)

Harsheel S Gowda — NNM23CS082 (Market Price Intelligence & Notifications)

UNDER THE GUIDANCE OF

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• RECAP — PHASE 1

What Was Already Delivered

Phase 1 shipped the core, explainable crop-recommendation engine end to end: a trained Random Forest classifier over real soil + weather data, SHAP-based explanations, and a 4-model comparison lab — verified against a running system, not a mockup.

99.3% Random Forest test accuracy on the 22-crop recommendation engine

SHAP Real per-feature explanation for every "why this crop?" recommendation

4 Algorithms honestly compared (RF, Gradient Boosting, XGBoost, SVM)

THIS PRESENTATION COVERS WHAT'S BEEN BUILT SINCE

Eight new modules layered on top of that Phase 1 core — four already committed (Yield, Profit, Water, Risk, Disease Detection) and three in active development (Fertilizer Advisor, Market Price Intelligence, Notifications) — all following the same rule: **every number shown is real, trained, or from a published source — never fabricated.**

• PRESENTATION PLAN

8 Modules, One per Member Pair

Same format as the Phase 1 literature review — each member owns two modules end to end: the problem it solves, the method/data behind it, and the honest result.

MEMBER 1

Alden Pais

NNM23CS020

→ Yield Prediction (ML)

→ Profit Prediction

MEMBER 2

Disha Shetty

NNM23CS067

→ Water Requirement

→ Yield Risk (ML)

MEMBER 3

Divya R

NNM23CS068

→ Disease Detection / Crop Doctor (ML)

→ Fertilizer Advisor

MEMBER 4

Harsheel S Gowda

NNM23CS082

→ Market Price Intelligence

→ Personalized Notifications

Yield Prediction

Estimates expected yield (tonnes/hectare) for a crop from its state, season and the trained regression model — a regional-historical estimate, not a per-farm forecast, and honestly labeled as such.

Dataset	325,000 district-level India crop-production records, 1997–2020
Active model	XGBoost Regressor (also trained: Random Forest, Gradient Boosting)
Coverage	11 of 22 crops — the rest honestly report "not available"
Shown	Right under the Top-5 crop recommendation on the detail page

MODEL PERFORMANCE

Algorithm	R^2	MAE
XGBoost (active)	0.919	1.03 t/ha
Random Forest	0.919	1.03 t/ha
Gradient Boosting	0.793	2.09 t/ha

HONEST LIMITATION

No access to an individual farm's soil, irrigation or this season's real weather — it answers "how has this crop performed in this state/season historically," not a per-farm prediction.

PRESENTED BY ALDEN PAIS

Profit Prediction

Not a trained model — a transparent calculation: real yield estimate × real historical market price, minus a cost the farmer supplies. Deliberately not modeled, because no honest per-farm cost-of-cultivation dataset exists.

Yield Estimate (t/ha) × 2016 Agmarknet Median Price = Revenue − Farmer-Supplied Cost = Estimated Profit

WHY COST ISN'T MODELED

No honest public per-farm cost-of-cultivation dataset was found. Inventing one would violate this project's own "never fabricate a number" rule — so cost stays farmer-supplied, by design.

COVERAGE

6 crops — the intersection of yield-supported crops and price-supported crops. Revenue uses real but non-inflation-adjusted 2016 mandi prices.

PRESENTED BY DISHA SHETTY

Water Requirement

Not a trained model — the real, published FAO formula, fetched and parsed directly from the primary FAO documents rather than recalled from memory.

THE FORMULA

```
ETo = p × (0.46 × Tmean + 8)  →  × FAO-56 Kc_mid  −  USDA effective rainfall
 =  Net Irrigation Need
```

Coverage	17 of 22 crops
Output	mm/day and liters/day/acre

A BUG CAUGHT BEFORE SHIPPING

An earlier version derived *p* from solar day-length instead of using FAO's own table — it produced an unrealistic ETo ≈ 22.8 mm/day vs. the real ≈ 6.2 mm/day for a test case. Caught, documented, and replaced with FAO's actual source table.

PRESENTED BY DISHA SHETTY

Yield Risk (ML-Based)

A trained classifier predicting LOW / MEDIUM / HIGH — how a crop has historically ranked nationally for a given state/season. Augments, never replaces, the existing rule-based risk dashboard.

Model	Random Forest, 3 algorithms compared
Test accuracy	55.3% on 2016–2020 (years unseen in training) vs. 33.3% baseline
Sanity check	Punjab correctly predicted LOW risk for rice — India's best-known high-yield rice state

A TRAIN/TEST LEAKAGE BUG, CAUGHT

A first random-split attempt reported a flattering **64.5%**. Proper cross-validation on the same data came back at only **36.9%** — near-duplicate (state, crop, season) rows from different years were leaking across train/test. Fixed with a temporal split; real result: 55.3%.

PRESENTED BY DIVYA R

Disease Detection — "AI Crop Doctor"

Upload a leaf photo → real diagnosis. A frozen ImageNet-pretrained MobileNetV3-Small extracts a 576-dim feature vector; a trained classifier head identifies crop + disease/healthy condition.

Dataset	3,875 real leaf images (PlantVillage-Dataset)
Coverage	13 classes across 4 crops — apple, maize, grapes, orange
Active model	Logistic Regression on MobileNetV3 features
Follow-up	Real curated care guidance (pathogen, cultural + chemical control class) per diagnosed class

MODEL PERFORMANCE

Algorithm	Accuracy
Logistic Regression (active)	97.94%
Gradient Boosting	95.74%
Random Forest	95.48%

VERIFIED LIVE

Maize Common Rust: 99.8% confidence. Healthy apple leaf: 99.98% confidence — both through the real `/api/predict/disease` endpoint, not just offline metrics.

• MODULE · IN PROGRESS

PRESENTED BY DIVYA R

Fertilizer Advisor

Rule-based decision support built on the same N/P/K/pH reference bands already used by the soil-health score — one source of truth, no drifting thresholds.

- Flags each nutrient DEFICIENT / ADEQUATE / EXCESS against agronomic reference ranges
- pH guidance for strongly acidic / alkaline soils (nutrient availability effects)
- Organic carbon (SOC) guidance — low / moderate / healthy bands
- Growth-stage-aware application timing: seedling → vegetative → flowering → fruiting → maturity
- Warns on excess nutrients and low soil moisture before application

DELIBERATELY NO DOSAGE NUMBER

Never outputs a kg/acre figure. Exact rates depend on a certified local soil-test report, crop variety, target yield and state agricultural-university guidance this system doesn't have — every response carries an explicit disclaimer.

• MODULE · IN PROGRESS

PRESENTED BY HARSHEEL S GOWDA

Market Price Intelligence

Real Agmarknet (Government of India) mandi price records, queried through a swappable `MarketPriceProvider` interface — mirrors the same pattern already used for weather, so a live Agmarknet/eNAM feed can be swapped in later without touching routes or frontend.

Records	263,520 real daily mandi price rows, calendar year 2016
Coverage	6 crops across up to 23 states and 848 markets/crop
Capabilities	Latest snapshot, price trend, cross-market comparison

SAMPLE DATA SCALE

Crop	Records	Markets
Rice	69,488	469
Maize	55,902	848
Pigeon peas	41,240	679

HONEST LABEL

Real government data, but a fixed historical year — clearly marked "not live" everywhere it's shown.

• MODULE · IN PROGRESS

PRESENTED BY HARSHEEL S GOWDA

Personalized Notifications

Derives alerts from the same real data every other module already computes — soil health, weather, risk, latest disease diagnosis, market price trend. Never invents or randomizes content.

Soil Health | Weather | Risk | Disease Diagnosis | Market Trend → Deduplicated Farmer Alert

5 ALERT CATEGORIES

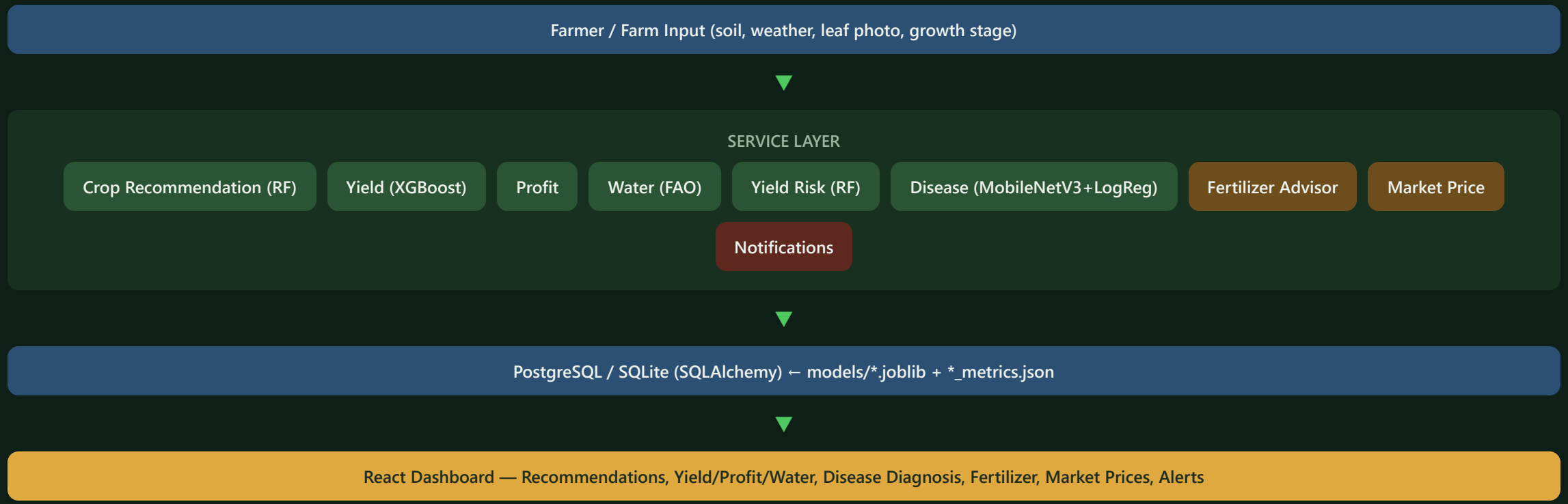
FERTILIZER (low soil score) · WEATHER (rain in forecast) · RISK (medium/high overall) · DISEASE (recent positive diagnosis) · MARKET (±5% price swing on the farmer's own top-recommended crop)

NO SPAM, BY DESIGN

A 12-hour dedup window skips re-creating an identical title+message, so repeated dashboard loads don't flood the farmer with duplicates.

• SYSTEM ARCHITECTURE

Updated End-to-End Block Diagram



Every Module's Real Number, Side by Side

Module	Method	Real Result	Crops Covered
Crop Recommendation	Random Forest	99.32% test accuracy	22
Yield Prediction	XGBoost Regressor	R^2 = 0.919, MAE = 1.03 t/ha	11
Profit Prediction	Revenue − Cost formula	Real 2016 mandi revenue	6
Water Requirement	FAO Blaney-Criddle	Real published formula + tables	17
Yield Risk	Random Forest classifier	55.3% test accuracy (33.3% baseline)	11
Disease Detection	MobileNetV3 + Logistic Regression	97.94% test accuracy	4 crops / 13 classes
Fertilizer Advisor	Rule-based reference bands	Same thresholds as soil health score	All
Market Price Intelligence	Real Agmarknet DB query	263,520 real mandi price records	6
Notifications	Rule-based aggregation	5 categories, 12h dedup window	All

• LIMITATIONS

Honest Constraints of the Current Build

YIELD / PROFIT / RISK

Regional-historical, not per-farm; profit's cost side is farmer-supplied, not modeled; risk covers only 11 of 22 crops.

DISEASE DETECTION

Frozen-feature-extractor, not fine-tuned; only 4 crops; PlantVillage's lab-condition images, not India field-condition photos.

WATER REQUIREMENT

Uses FAO's own "rough estimate" Blaney-Criddle method (not the more accurate Penman-Monteith); state-capital latitude, not farm GPS.

FERTILIZER / MARKET / NOTIFICATIONS

Fertilizer gives no dosage number by design; market prices are a fixed 2016 snapshot, not live; notifications only cover the signals already computed elsewhere.

• CURRENT IMPLEMENTATION

Overall Project Progress



Crop recommendation, yield, profit, water requirement, yield risk and disease detection are committed and verified end-to-end. Fertilizer advisor, market price intelligence and notifications are functionally complete and in final integration/testing.

STILL NOT IMPLEMENTED

Real LLM/RAG-backed AI assistant, voice interface, multilingual support, satellite/NDVI monitoring, IoT sensor ingestion — all have UI shells or roadmap placeholders in place already.

NEXT STEPS

Finish integration testing for the three in-progress modules, then move to an automated backend/frontend test suite and CI pipeline.

• CONCLUSION

Summary & Path Forward

Phase 1 Recommendation	Yield & Profit	Water & Risk	Disease Detection	Fertilizer	Market Price	Notifications	Future Scope

AgriAI 360 has grown from a single explainable recommendation engine into an eight-module farm decision-support platform — each module trained, formula-verified, or built on real government/scientific data, and each one honest about what it doesn't yet know.

REFERENCES

Phase 1's 20-paper literature review (unchanged) + project documentation (docs/dataset.md, docs/ml-methodology.md, docs/architecture.md) + each module's own train.py / service file + models/*_metrics.json + models/market_price_report.json.